Detecting and Mitigating Adversarial Machine Learning Attacks in Autonomous Vehicles Within the Internet of Vehicles

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***Abstract*—Adversarial Machine Learning (AML), particularly model poisoning, presents a critical threat to Autonomous Vehicles (AVs) in the Internet of Vehicles (IoV) environment. To address this challenge, we propose a framework that integrates Federated Learning (FL) with a Deep Learning-based Intrusion Detection and Behavior Monitoring (DL-based IBM) system, a vehicle scoring mechanism, Digital Twin (DT), and Generative AI (Gen-AI) to enhance security in IoV environments. Each AV employs a DL-based IBM as its local model, which is trained on vehicle-local operational data and contributes to a global model through FL aggregation. A vehicle scoring system is responsible for identifying and flagging compromised AVs (Zombies) exhibiting suspicious behavior. When the DT detects that the Zombie's model has been manipulated through poisoning attacks, the DT updates the compromised model with the correct global model to restore normal operation. Furthermore, DT leverages Gen-AI to simulate and learn from novel attack scenarios that AVs have not previously encountered, ensuring that the framework adapts to evolving threats. The simulation results demonstrate significant improvements in resilience and detection accuracy, with F1 scores reaching 99% for known attacks and exceeding 87% for new unseen threats. This integrated approach ensures robust and adaptive protection for AVs, maintaining high trust and performance in the dynamic and adversarial IoV network.**

***Index Terms*—Autonomous vehicle, adversarial machine learning, digital twin, federated learning, Gen-AI, deep learning.**

I. Introduction

THE Internet of Vehicles (IoV) represents a rapidly evolving network in which vehicles, road infrastructure, and other systems communicate and collaborate to improve traffic management, safety, and driving efficiency. IoV enables the exchange of real-time data between Vehicles (V2V), Vehicles and Infrastructure (V2I) and Vehicles and other devices (V2X). This vast amount of data enables intelligent decisions about vehicle behavior, route optimization, and accident prevention. Autonomous Vehicles (AVs) rely on IoV for safe and efficient navigation. By processing real-time information about road conditions, other vehicles, and potential hazards, IoV provides critical input to AV decision-making systems [1], [2], [3].

To manage this complex network of interactions, Machine Learning (ML) and Deep Learning (DL) play a crucial role in IoV. ML/DL algorithms enable vehicles to learn from large datasets, process sensor data, and make predictions about their environment [4]. Applications of ML/DL in vehicles include intrusion detection systems to identify malicious activities, anomaly detection to identify unsafe driving behaviors, and route planning optimization to ensure smooth traffic flow [5], [6], [7], [8], [9]. For example, [10] proposes a Decision Tree–based IDS is proposed using in-depth CAN traffic analysis to improve detection robustness in automotive networks. Similarly, [11] presents an ML-driven ECU fingerprinting system using SVM and ANN models for embedded intrusion detection in vehicles. By leveraging ML/DL, IoV systems can improve the safety, efficiency, and adaptability of the network, enabling AVs to respond to real-world conditions dynamically and intelligently.

One of the key challenges in IoV is the need to continuously update and improve ML models without compromising data privacy. Vehicles collect sensitive data, including location, speed, driving habits, and interactions with other vehicles. Sharing these data with centralized servers for training ML models poses privacy concerns and, in some cases, the sheer volume of data makes it impractical to transmit.

Federated Learning (FL) addresses this challenge by allowing vehicles to collaboratively train ML models without sharing their raw data [12]. In IoV, FL enables AVs to improve their performance in tasks such as intrusion detection, behavior monitoring, and anomaly detection [13]. Since vehicles operate in diverse environments and encounter unique challenges, FL allows the global model to be enhanced with data from different driving scenarios. This collaborative learning process makes the IoV system more adaptable to changing conditions and more resilient against potential threats [13], [14], [15].

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However, FL frameworks such as horizontal FL, vertical FL and federated transfer learning [16], face significant security challenges, especially from Adversarial Machine Learning (AML) attacks [17], which can compromise both local and global model performance. In FL, AML attacks exploit the collaborative and decentralized nature of training, targeting local training or global aggregation phases to introduce vulnerabilities into the system [18].

Model poisoning attacks are a highly disruptive form of AML, in which a malicious participant manipulates the local model updates, such as gradients or weights, prior to aggregation in the federated learning process. The attacker's primary objective is to influence the global model's learning trajectory, often by injecting subtle but targeted deviations that accumulate over time. This manipulation can lead to degraded model performance or, in more severe cases, cause the global model to misclassify critical input, such as failing to detect behavioral anomalies in safety-critical applications like autonomous vehicles. These attacks are particularly insidious because they can remain undetected during training, thus exploiting the decentralized trust model inherent in FL systems [19], [20].

Digital Twin (DT) technology-a virtual replica of a physical system such as an autonomous vehicle-can help detect and mitigate AML attacks in FL [21].The DT models key subsystems, including intrusion detection, anomaly monitoring, communication modules, and sensors, within a synchronized virtual environment. By mirroring the operational state of the physical vehicle, the DT can simulate realistic driving and attack scenarios, evaluate model responses, and validate the integrity of FL-trained models in real time. This continuous simulation and evaluation capability improves the resilience of the system by enabling rapid detection, diagnosis, and mitigation of adversarial manipulation before it affects real-world operations [22], [23], [24], [25], [26].

Furthermore, DTs use synthetic adversarial data generated by Generative AI (Gen-AI) to create Digital Generative Twins (DGTs), which improve local model training with a variety of attack patterns and changes, increasing adaptability to new AML threats. This ongoing adversarial training strengthens the robustness of the model and supports a better detection of dynamic anomalies.

Motivation. There is an urgent need to address the growing threat of AML attacks on AVs in next-generation IoV networks. As AVs increasingly rely on ML and DL models for decision making, perception, and coordination, attacks such as model poisoning can seriously compromise their safety and performance. The widespread use and dynamic nature of IoV environments make them more vulnerable to these threats. This highlights the need for secure, adaptive, and reliable learning systems that can protect AVs from malicious behavior and maintain reliable operation under adversarial conditions.

Contributions. Motivated by this need, this work presents several key contributions aimed at enhancing the security and performance of AVs in IoV networks. As a case study (see Section II), an Intrusion detection and anomaly Behavior Monitoring (IBM) system, serving as a security component of the AV, is developed and evaluated against AML attacks, specifically model poisoning attacks, to assess its resilience within a FL framework. The primary focus is to examine how model poisoning attacks affect the IBM system and how these attacks can be addressed by leveraging DGT. The main contributions are as follows:

- Proposing a system that enables AVs to collaboratively improve their IBM system using FL.
- Mitigating the risks posed by AML attacks by integrating DT technology, which simulates vehicle behaviors and detects model poisoning attack in a controlled environment, helping to safeguard the global model.
- Generating synthetic data by Gen-AI enhances the DT's knowledge base, helping to detect unknown anomalies and emerging threats not seen in real-world data, providing an adaptive defense mechanism.

Assumption. In this study, we assume that secure communication is established between all critical components of the system, including AVs and the Cloud, AVs and DTs, AVs and Roadside Units (RSUs), and between DTs and the Cloud. To ensure confidentiality, integrity, and authentication, all communication channels leverage lightweight encryption and authentication mechanisms that comply with the principles of Zero Trust Architecture (ZTA). The DT is deployed on an isolated virtual machine (VM) using isolated deployment architectures, ensuring strict separation from the AV and protection against external interference or compromise. However, we acknowledge that in large-scale IoV environments, bandwidth and latency constraints can significantly affect real-time data transfer and synchronization between AVs, RSUs, and DTs. Therefore, the DT and FL aggregator can be deployed at the edge layer, such as RSUs, to minimize communication latency and reduce dependency on centralized cloud resources. This distributed design helps alleviate bandwidth congestion, ensures faster feedback loops for AV decision-making, and enhances scalability and reliability in dynamic IoV environments.

The structure of this paper is as follows: Section II explores a case study focusing on intrusion detection and behavior monitoring within IoV systems. Section III evaluates the effectiveness of the proposed defense mechanism through formal verification. Section IV outlines the experimental scenarios and setup, followed by a discussion of the results. Finally, Section V summarizes the key findings and outlines the conclusions and directions for future work.

II. Case Study: Intrusion Detection and Anomaly Behavior Monitoring in Autonomous Vehicles Under Model Poisoning Attacks

This case study focuses on improving the security and safety of AVs in the IoV network using an IBM system as a representative example of existing ML/DL-based applications deployed in autonomous vehicles. The IBM system is used to examine the impact of model poisoning attacks in a FL setting and to evaluate mitigation strategies. To address these attacks, the framework uses a real-time vehicle behavior scoring system and DGT technology, enabling the detection, validation, and recovery of compromised models. Using secure

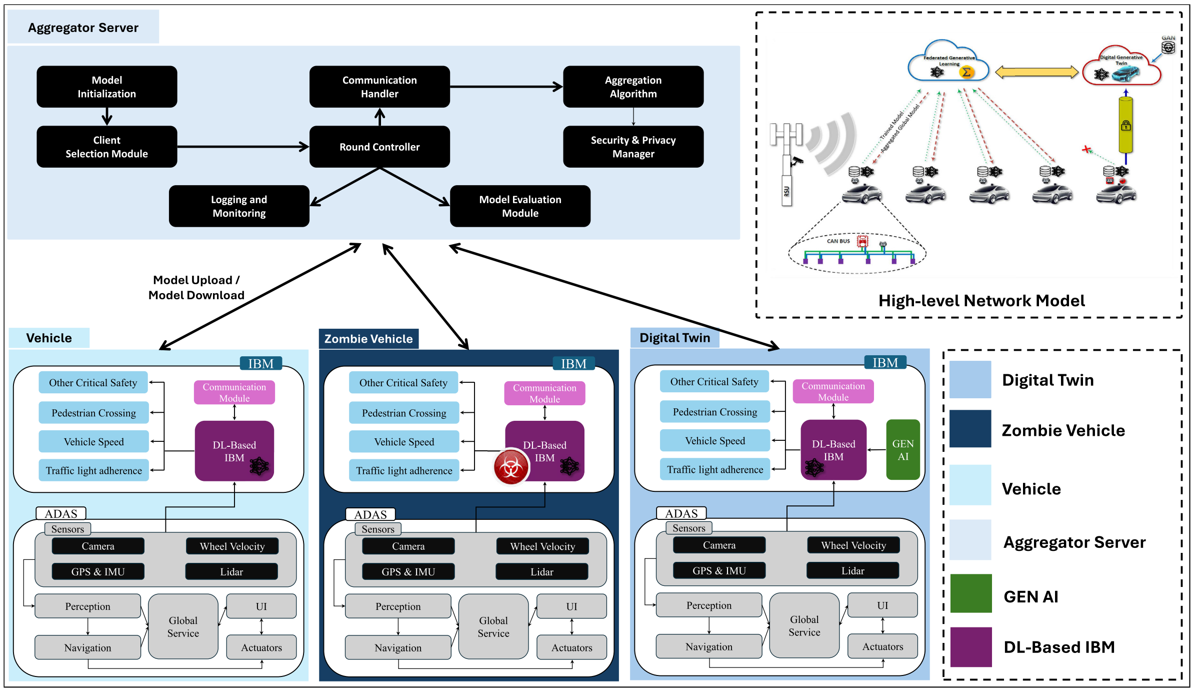

Fig. 1. Proposed framework.

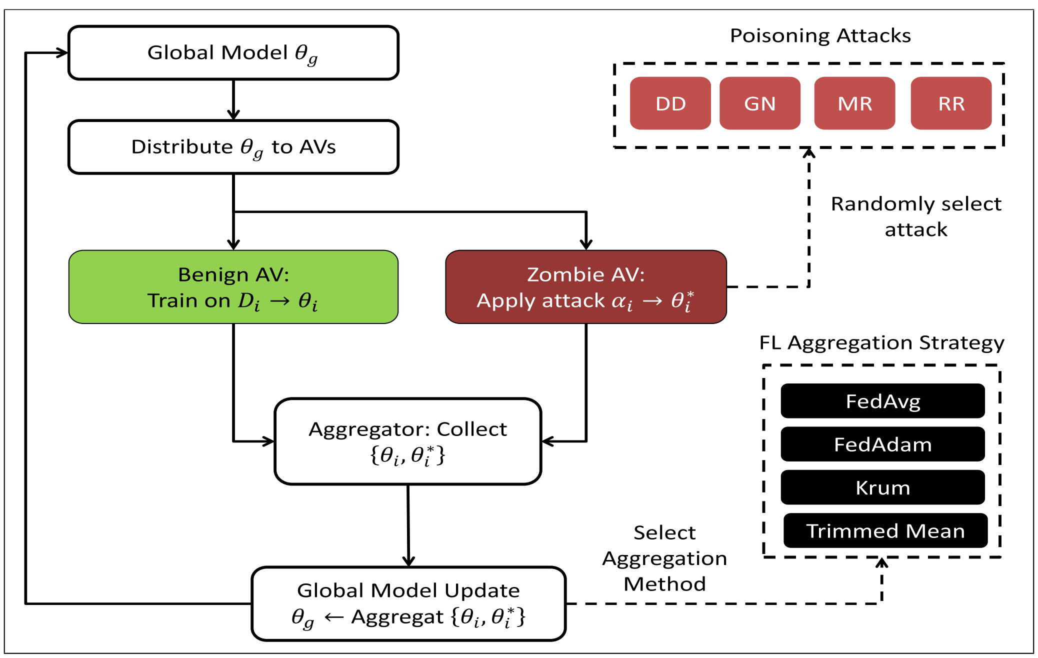

Fig. 2. Adversarial manipulation scenarios in FL with four attack types: GN, DD, MR, and RR.

LTE / 5G communication, cloud-based aggregation, and RSU, the system ensures continuous monitoring and provides an adaptive defense against evolving AML threats in connected AVs.

A. System Model

The system model designed to improve the security and safety of AVs in the IoV network is made up of several interconnected components. These components work together to form a robust framework for detecting and mitigating anomalous behaviors, as well as addressing potential cybersecurity threats such as model poisoning attacks. As shown in Fig. 2, this system is implemented within a network of AVs, RSUs, cloud infrastructure, and DTs located at the edge of the network, all connected through secure LTE/5G communication channels.

Each AV is equipped with an DL-based IBM system, which serves as the local model within the FL framework. This system continuously monitors critical vehicle behaviors, such as traffic light adherence, pedestrian lane crossings, vehicle speed, sudden braking, and other safety-critical driving behaviors.

In the context of autonomous vehicles, the Controller Area Network (CAN) protocol is utilized for communication between the vehicle's Electronic Control Units (ECUs). The DL-based IBM system is integrated into this CAN bus, which functions as an ECU to seamlessly monitor and analyze real-time data for intrusion detection and anomaly behavior monitoring. This ensures that the security system operates within the vehicle's existing communication framework, enabling effective real-time detection of anomalies.

The cloud serves as the central aggregation point within the system, where local models from multiple vehicles are collected and aggregated to form a global model. This global model benefits from the diversity of data collected by individual vehicles, improving the system's ability to detect a wide range of abnormal behaviors and attacks across the network. However, the system also considers the possibility of AML attacks, particularly model poisoning attacks, where malicious vehicles may inject compromised models into the aggregation process to degrade the global model's performance.

To address these potential threats, the RSUs and other components of the network infrastructure continuously evaluate the behavior of each AV. Each vehicle is assigned a behavior score based on its performance in real-world driving conditions, such as adherence to safety standards and proper handling of critical situations. Vehicles with low scores, which may indicate potential AML attacks or anomalies, are flagged for further analysis. These low-score vehicles undergo additional testing in the DT environment.

The DT acts as a virtual testing and simulation environment for vehicles that exhibit abnormal behavior or low performance scores. In this controlled environment, the vehicle's local model is subjected to a series of tests, including the detection of potential model poisoning attacks or other cyber-security threats. By incorporating real-time and synthetic data generated by Gen-AI, DT can also identify unknown attack vectors and evolving threats, ensuring that anomaly detection models remain resilient even in the presence of advanced and emerging threats.

Secure communication is essential for system operation, and all interactions between vehicles, cloud, RSUs, and the DT system are conducted over a secure LTE/5G network. This ensures low-latency, high-speed data transmission for real-time monitoring and anomaly detection. The communication framework incorporates encryption and authentication protocols to protect sensitive data and prevent unauthorized access, ensuring the integrity of the system.

This system provides a comprehensive framework for securing AVs in the IoV network. By integrating DL-based intrusion detection, anomaly behavior monitoring, cloud-based aggregation, and DT testing, the system enhances real-time detection capabilities while mitigating the risks posed by model poisoning attacks and other cyber-security threats.

B. AML Attacks: Model Poisoning

Model poisoning attacks are a subset of AML techniques that specifically target FL systems. In these attacks, adversaries aim to corrupt the model by altering the local model updates before they are aggregated into the global model. This can degrade the performance of the global model, leading to false positives or missing detection in systems.

A typical FL setup involves a collection of local models θ_i, each trained in a private data partition D_i located on the autonomous vehicle AV_i, where $i = 1, 2, \ldots, N$. After local training, these models share their weight updates $\Delta\theta_i$ with the central aggregator (cloud), which aggregates them into a global model, denoted θ_g, according to:

$$\theta_g^{(t+1)} = \theta_g^{(t)} + \frac{1}{N} \sum_{i=1}^{N} \Delta\theta_i^{(t)} \tag{1}$$

Generally model poisoning attacks manipulate the local model parameters θ_i so that the updates $\Delta\theta_i$ become poisoned $\Delta\theta_i^*$, with the objective of negatively influencing the global model.

For a benign update, the local training loss function can be represented as

$$\mathcal{L}_i(\theta) = \frac{1}{|D_i|} \sum_{(x,y)\in D_i} l\left(f_\theta\left(x\right), y\right) \tag{2}$$

$$\Delta\theta_i^* = \Delta\theta_i + \delta_i \tag{3}$$

where $f_\theta(x)$ is the prediction of the model with parameters θ, $l(f_\theta(x), y)$ is the loss function, and D_i is the local dataset of vehicle i, δ_i is the adversarial perturbation introduced by the attacker, which varies depending on the specific attack vector. The goal of δ_i is to maximize global loss or to specifically degrade detection performance in a subset of malicious activities, such as avoiding detection of specific intrusions.

When poisoned updates are aggregated with benign updates, the global model becomes compromised. The new global model update after including the poisoned model is:

$$\theta_g^{(t+1)} = \theta_g^{(t)} + \frac{1}{N} \left(\sum_{i=1}^{N-1} \Delta\theta_i + \Delta\theta_i^* \right) \tag{4}$$

In our poisoning attack scenarios, we simulate four different types of adversarial manipulation: Gaussian noise (GN), directional drift (DD), model reversal (MR) and random replacement (RR). These attacks represent common strategies used to compromise FL systems by corrupting local model updates prior to aggregation. Gaussian noise introduces random perturbations to model parameters, while directional drift systematically shifts updates in a specific direction to bias the global model. Model reversal inverts the gradients to counteract the learning process, and random replacement substitutes legitimate models with completely unrelated or malicious ones. Each of these techniques aims to degrade the performance or manipulate the behavior of the global model, thereby undermining the reliability of autonomous vehicle decision making in IoV environments. Algorithm 1 illustrates the simulation of AML in FL, which incorporates four poisoning strategies of the model. In this algorithm, a zombie refers to a vehicle compromised by a model poisoning attack. It is identified through our vehicle scoring system, which will be discussed in Section II-D.

Algorithm 1 AML Simulation in FL

Input: Global model θ_g, Local datasets $\{D_1, D_2, \ldots, D_N\}$, Total vehicles N, Simulation rounds R, Number of zombie vehicles Z, Attack types $\mathcal{A}$ = {Gaussian, Drift, Reversal, Replacement}, μ is drift intensity
Output: Poisoned local models $\{\theta_i^*\}$, Final global model θ_g^{final}
Initialize: $\theta_g \leftarrow$ random initialization
for each round $r = 1$ to R **do**
  Server distributes θ_g to all vehicles
  for each vehicle $i \in \{1, 2, \ldots, N\}$ **in parallel do**
    if vehicle i is a zombie ($i \leq Z$) **then**
      Choose attack type $a_i \in \mathcal{A}$
      Apply attack based on a_i**:**
      if a_i = Gaussian **then**
        Sample noise $\eta \sim \mathcal{N}(0, \sigma^2)$
        $\theta_i^* \leftarrow \theta_g + \eta$
      else if a_i = Drift **then**
      Generate drift vector d toward malicious direction
        $\theta_i^* \leftarrow \theta_g + \mu d$
      else if a_i = Reversal **then**
        Train local model θ_i on D_i with θ_g
        Estimate $\nabla \mathcal{L}_i$ from model
        Inverted gradient $\theta_i^* \leftarrow \theta_g - \alpha \nabla \mathcal{L}_i$
      else if a_i = Replacement **then**
        $\theta_i^* \leftarrow$ random model parameters
      end if
    else
      Train local model θ_i^* on D_i using θ_g
    end if
    Send θ_i^* to the aggregator
  end for
  Aggregator updates global model $\theta_g \leftarrow \text{Aggregate}(\{\theta_i^*\})$
end for
return Final global model θ_g^{final}, $\{\theta_i^*\}$

C. DL-Based IBM System

The DL-based IBM system is designed to improve the security and safety of AVs in IoV. The primary function of this system is to detect both intrusions (external cyber-security threats) and anomalous behavior (unusual or unsafe driving patterns) in real-time. The DL-based IBM system operates as a local model onboard each vehicle, utilizing data from various sensors and communication protocols, and is periodically updated using federated learning.

The DL-based IBM system takes as input a stream of data from the vehicle's sensors and internal communication bus, such as vehicle speed, GPS position, Traffic light adherence, Pedestrian lane crossings, Sudden braking events, CAN bus data for communication between ECUs, and network traffic data for detecting cyber-security threats. Each feature in the input data can be represented as a multidimensional vector:

$$X_t = \left[x_1^{(t)}, x_2^{(t)}, \cdots, x_n^{(t)}\right] \tag{5}$$

where X_t is the input feature vector at time t, and n is the number of features.

The DL-based IBM system formulates intrusion and anomaly detection as a classification task, with the aim of labeling each observed behavior or network traffic instance as normal or abnormal. This can be formalized as a binary classification task, where the output of the system is a probability distribution over two classes:

$$P(y|X_t) = \text{softmax}\left(f_\phi(X_t)\right) \tag{6}$$

where $P(y|X_t)$ represents the probability of the class y (normal or anomalous behavior) given the input X_t, f_ϕ is the deep learning model parameterized by ϕ, softmax(.) is the softmax function used to convert the output of the model into a probability distribution.

The system is trained to minimize the binary cross-entropy loss function:

$$\mathcal{G}(\phi) = -\frac{1}{m}\sum_{i=1}^{m} \left[y_i \log P(y_i|X_i) + (1 - y_i) \log (1 - P(y_i|X_i))\right] \tag{7}$$

where m is the number of training samples, y_i is the true label of the i-th sample (1 for anomalous, 0 for normal), $P(y_i|X_i)$ is the predicted probability of the i-th sample belonging to class y_i.

D. Vehicle Scoring System (VSS)

To evaluate the safety and performance of AVs, a behavior-based scoring system is proposed that continuously monitors driving actions and assigns scores based on key performance metrics. Our system integrates both internal (on-board system data) and external (infrastructure-based observations) factors to provide a comprehensive assessment of vehicle behavior. The scoring is based on parameters such as the ability to maintain a safe distance from other vehicles under varying traffic conditions, adherence to speed limits with adaptive control in different types of road and environmental conditions, and effective handling of critical situations involving sudden braking, obstacle avoidance, or interactions with pedestrians. Although the present work focuses on logical-level indicators relevant to AML detection, low-level physical attacks such as CAN spoofing and ECU tampering [27] could also be addressed in future extensions through the integration of physical fingerprinting techniques, which are beyond the scope of this study.

1) Internal Scoring System (ISS): The proposed ISS continuously monitors the behavior of the vehicle based on sensor data, such as radar, LiDAR, and cameras. This system evaluates key performance metrics, including safety distance, speed control, and handling of critical situations. The internal score $S^{in}(t)$ of AV i at time t is a weighted sum of various behavior metrics.

$$S_i^{in}(t) = w_1 \times D_{i,safe}(t) + w_2 \times V_{i,opt}(t) + w_3 \times H_{i,crit}(t) \tag{8}$$

where $D_{i,safe}$ is a score related to the distance of the AV i from other surrounding AVs. This score decreases if the vehicle is dangerously close to other vehicles, V_{opt} is the optimal speed score, where speed control is evaluated based on the difference between vehicle speed and the recommended speed for the segment or situation of the road, and H_{crit} represents the handling score for critical situations, where the vehicle's ability to react appropriately in sudden or dangerous situations is evaluated. w_1, w_2, and w_3 are weighting factors that reflect the relative importance of each metric. Each of these metrics is normalized between 0 and 1, with higher values indicating better performance.

The safety distance metric $D_{safe}(t)$ can be modeled as a function of current speed $v(t)$ and distance to the closest vehicle $d(t)$:

$$D_{safe}(t) = 1 - \exp\left(-\frac{d(t)}{v(t)+\epsilon}\right) \tag{9}$$

where ϵ is a small positive constant to avoid division by zero. This formula ensures that the score decreases as the vehicle approaches others too quickly relative to its speed.

The optimal speed metric $V_{opt}(t)$ measures the deviation from the optimal speed $v_{opt}(t)$ for the given road segment. It can be expressed as:

$$V_{opt}(t) = 1 - \left|\frac{v(t) - v_{opt}(t)}{v_{opt}(t)}\right| \tag{10}$$

here, $v(t)$ is the actual speed of the vehicle at time t, $v_{opt}(t)$ is the speed limit or the recommended speed for the type of road and traffic conditions.

The handling in critical situations score $H_{crit}(t)$ is based on how well the vehicle reacts to emergencies, such as sudden braking or obstacle avoidance. It can be represented by a binary or probabilistic function that assigns a high score if the vehicle successfully avoids collisions and maintains control:

$$H_{crit}(t) = P(\text{successful response} \mid \text{critical event}) \tag{11}$$

This probability can be learned from historical data or expert rules, with high scores for successful reactions.

2) External Scoring System (ESS): The proposed ESS uses road infrastructure, such as traffic cameras, speed detectors, and connected sensors, to assess vehicle behavior from an external perspective. The external score $S^{ex}(t)$ of AV i is calculated in a similar manner, but with a focus on factors that the infrastructure can measure, such as adherence to traffic signals, speed limits, and lane discipline:

$$S_i^{ex}(t) = w_4 \times T_{i,signal}(t) + w_5 \times L_{i,lane}(t) + w_6 \times V_{i,comp}(t) \tag{12}$$

where T_{signal} is the traffic signal adherence score, which is 1 if the vehicle obeys traffic signals and 0 otherwise. L_{lane} is the lane discipline score, which evaluates how well the vehicle stays in the lanes and makes the appropriate lane changes. V_{comp} is the speed compliance score that measures how well the vehicle complies with the speed limits enforced by the road infrastructure.

3) Combined Scoring System: The final score $S(t)$ of AV i at time t is a weighted combination of internal and external scores:

$$S_i(t) = \alpha \times S_i^{in}(t) + \beta \times S_i^{ex}(t) \tag{13}$$

where α and β are weighting factors that adjust the relative contribution of the internal and external systems to the

Algorithm 2 Anomaly Detection and Model Recovery in DL-Based IBM Using Digital Twin

Require: DL-based IBM on vehicle: θ_v, ISS on vehicle: S_v^{in}, ESS on RSU: S_v^{ex}, Threshold δ, Digital Twin IBM: θ_{dt}

Monitor driving behaviors (traffic light, pedestrian lane, speed, braking, etc.) and collect input data x
Evaluate behavior scores using S_v^{in} and S_v^{ex}
Predict anomaly using $\theta_v(x)$
Compute score difference: $\Delta = \left|S_v^{in} - S_v^{ex}\right|$
if $\Delta > \varphi$ **then**
  Send x and θ_v to the Digital Twin
  Compute $\hat{y}_{dt} = \theta_{dt}(x)$
  Compute $\hat{y}_v = \theta_v(x)$
  if $|\hat{y}_{dt} - \hat{y}_v| > \varphi$ **then**
    θ_v is compromised
    Replace $\theta_v \leftarrow \theta_{dt}$
  else
    θ_v is safe
  end if
else
  Behavior normal, no action required
end if

overall score. The combined score provides a comprehensive evaluation of vehicle behavior by incorporating both on-board and external monitoring systems. Algorithm 2 illustrates the process of identifying a suspicious AV using the vehicle scoring system, followed by recovery of its compromised model through validation and replacement through DT.

E. Digital Generative Twins (DGT)

DGT combines the concepts of digital twins and Gen-AI to create virtual replicas of physical systems, processes, or environments. DT is cutting-edge technology that creates a dynamic, virtual replica of a physical system, enabling real-time monitoring, simulation, and analysis. This capability is particularly crucial for complex environments like IoV, where the coordination between autonomous vehicles, infrastructure, and central control systems requires precision and adaptability. Gen-AI complements DT technology by synthesizing data and generating scenarios that mimic real-world conditions [28]. Different types of Gen-AI tools include Generative Adversarial Networks (GANs), Variational Autoencoders (VAEs), transformer-based models such as GPT, and time series generators such as TimeGAN. When combined, DT and Gen-AI form a robust framework that not only enhances system performance but also strengthens security against sophisticated adversarial threats, including AML and model poisoning.

DT technology operates by establishing a digital representation of real-world objects or systems, linked through continuous data streams for synchronization. The essential components include the data acquisition layer, the simulation core, and the analysis and feedback loop. The data acquisition layer collects real-time data from IoV sensors, communication channels, and vehicle systems. The simulation core is a computing module that models and predicts the behavior of the system under various conditions using physical laws, learned behaviors, or a combination of both. The analysis and feedback loop integrates predictive analytics and anomaly detection algorithms to provide insights and corrective actions to the physical system. Mathematically, a DT can be represented as:

$$\mathbf{DT} = (X, \Theta, \Phi) \tag{14}$$

where X represents the state variables, such as vehicle position, speed, sensor readings, Θ denotes the physical and digital parameters of the system, and Φ represents the transfer functions or machine learning models mapping system behavior.

DGT is one of the effective solutions to address AML attacks. In this study, DGT is used to simulate model poisoning attacks, as shown in Algorithm 1, and to evaluate suspicious AVs by testing the accuracy of the ML/DL models deployed on them. This allows the system to train in advance and become more resilient against model poisoning. By exposing the learning framework to synthetic adversarial scenarios, the system can better recognize and defend against malicious patterns.

F. Generating Unknown Attack for IoV Network Using cGAN

In this study, Conditional GAN (cGAN) is used to generate sophisticated adversarial attack vectors that mimic realistic model manipulation. The cGAN consists of two neural networks, a generator and a discriminator, that are trained in opposition. The generator takes as input a clean model weight vector and a condition vector representing the type and strength of attack, and learns to output a plausible attack delta that perturbs the clean model. During the same time, the discriminator receives both real and generated attack deltas, along with the corresponding clean weights and conditions, and learns to distinguish between genuine and synthetic perturbations. Through this adversarial training process, the generator becomes increasingly skilled in producing attack vectors that are indistinguishable from real ones (Algorithm 3).

Algorithm 3 cGAN for Adversarial Attack Generation

Require: Clean weights W_c, real attack deltas Δ_a, condition vectors c, generator G, discriminator D, number of epochs N, batch size B

Initialize parameters of G and D
for epoch = 1 to N **do**
  for each mini-batch of size B **do**
    Sample W_c, Δ_a, and c from dataset
    Generate adversarial deltas $\Delta_g \leftarrow G(W_c, c)$
    Discriminator score for actual attack $s_a \leftarrow D(\Delta_a, W_c, c)$
    Discriminator score for generated attack $s_g \leftarrow D(\Delta_g, W_c, c)$
    Compute discriminator loss:

$$L_D = \frac{1}{2}\left(\mathrm{BCE}(s_a, 1) + \mathrm{BCE}(s_g, 0)\right)$$

    Update D using gradient $\nabla_D L_D$
    Re-generate for generator update $\Delta_g \leftarrow G(W_c, c)$
    $s_g \leftarrow D(\Delta_g, W_c, c)$
    Compute generator loss:

$$L_G = \mathrm{BCE}(s_g, 1)$$

    Update G using gradient $\nabla_G L_G$
  end for
end for
return Trained generator G

III. Latency-Accuracy Tradeoff Modeling in FL for IoV

In IoV environments, AV decision-making requires accurate and timely learning. In FL, the accuracy of the model is directly influenced by both computation and communication time, making it essential to design systems that optimize these factors jointly to maintain real-time performance and efficient use of resources. For example, delays in model synchronization between AVs and the edge server during critical maneuvers, such as lane-keeping corrections, steering control, or emergency braking, can result in outdated decisions or missed anomaly detections. Therefore, maintaining a balance between computational efficiency and low-latency communication is vital to enable AVs to respond accurately and quickly to dynamic driving conditions in IoV systems.

A. Computation Time

In each AV, the computation time is a critical factor that affects the overall latency in the learning process. The computation time required for AV i to complete the local training on D_i samples is given by the following:

$$T_{\text{comp},i} = \frac{C \cdot D_i}{f_i} \tag{15}$$

where C is the number of CPU cycles required to process one data sample, D_i refers to the number of local data samples in AV i, and f_i is the CPU frequency (cycles per second, in GHz) for AV i.

On the cloud side, computational time is primarily spent aggregating model updates. Assuming the cloud has high performance computing capabilities, the aggregation delay is relatively small but increases with the number of participating AVs N and model size S. The cloud-side aggregation time can be approximated as

$$T_{\text{agg}} = \alpha \cdot N \cdot S \tag{16}$$

where α is a proportionality constant that represents the aggregation cost per model element. The end-to-end computation time is thus:

$$T_{\text{total-comp}} = \max_i T_{\text{comp},i} + T_{\text{agg}} \tag{17}$$

This expression highlights that the slowest AV dominates the edge-side computation delay, and the total delay also includes cloud aggregation time. Reducing $T_{comp,i}$ (by increasing f_i, reducing D_i, or simplifying the model) is essential to ensure timely model updates and avoid delayed contributions, which negatively impact convergence and global model accuracy.

B. Communication Time

It includes the transmission delay (T_{tx}), the propagation delay (T_{prop}), the queue delay (T_{queue}), and the processing delay (T_{proc}). In this study, we neglect the processing time, as it is typically negligible compared to the transmission and queueing delays in FL communication.

$$T_{\text{comm},i} = T_{\text{tx},i} + T_{\text{prop},i} + T_{\text{queue},i} + T_{\text{proc},i} \quad (18)$$

In FL, communication occurs in two phases: download in which the cloud sends the global model to AVs and upload in which AVs send local model updates to the cloud. Therefore, the total communication time per round is as follows:

$$T_{\text{total-comm},i} = T^{d}_{\text{comm},i} + T^{u}_{\text{comm},i} \quad (19)$$

Each can be modeled using the same form, possibly with different model sizes (S^u and S^d) and bandwidth in the uplink (B^u) and downlink (B^d).

The transmission delay is determined by the size of the model and the communication bandwidth.

$$T_{\text{tx},i} = \frac{S_i}{B_i} \quad (20)$$

This term quantifies the time required to transmit the model update from the AV i to the aggregator. Here, S_i is the size of the local model update in bits, and B_i represents the available communication bandwidth in bits per second (bps).

The propagation delay represents the physical time required for a signal to travel from the AV to the cloud.

$$T_{\text{prop},i} = \frac{d_i}{v} \quad (21)$$

where d_i is the distance and v is the signal propagation speed in the medium (typically 3×10^8 m/s in free space). Although relatively small compared to other delays, the propagation delay becomes significant in scenarios where the cloud is geographically distant, making edge server placement a critical design decision.

The queue delay represents the time it takes for a message to wait in the queue before being processed/transmitted.

$$T_{\text{queue},i} = \frac{\rho_i}{\mu_i(1-\rho_i)} \quad (22)$$

This component models the delay due to network congestion and resource contention, where μ_i is the service rate (in bits/s or packets/s), and $\rho_i = \lambda_i/\mu_i$ is the utilization of the system with λ_i as the arrival rate. As the system approaches saturation ($\rho_i \to 1$), the queueing delay increases dramatically, highlighting the need for dynamic scheduling and congestion control mechanisms.

As explained above, in FL, both communication and computation efficiency critically influence the accuracy and performance of the local/global model. Communication delays, caused by large model sizes, limited bandwidth, or long transmission distances, can result in delayed updates that fail to capture the current state of the environment, thus reducing the accuracy of the model. On the computation side, constrained resources in AVs can lead to delayed or too early stopped local training, resulting in low-quality updates. Furthermore, aggregation delays in the cloud, due to the heterogeneity and complexity of the AV model, weaken the synchronization between the AVs, slowing the convergence. Together, these inefficiencies hinder the timely adaptation to dynamic road conditions and reduce the overall effectiveness of the FL system. To ensure high model accuracy while respecting system constraints, it is necessary to jointly optimize communication and computation. An approach is to formulate accuracy A as a function $A = \mathcal{F}(S, B, d, f_i, D_i)$, and to define an optimization problem that considers both communication and computation delays.

$$\begin{aligned} \textbf{P:} \quad & \max_{\{S,B,d,f_i,D_i,\theta_i\}} \quad A(S, B, d, f_i, D_i) \\ \text{s.t.} \quad & \underbrace{T_{\text{total-comp},i}}_{\text{computation time}} + \underbrace{T_{\text{total-comm},i}}_{\text{communication time}} \leq \underbrace{\tau_{max}}_{\text{latency constraint}} \end{aligned} \quad (23)$$

This formulation enables a multidimensional trade-off. Increasing the size of the model can improve the accuracy, but only if sufficient bandwidth and computational resources are available to meet the latency constraint τ_{max}. Optimizing this balance is critical for high-accuracy and real-time learning in IoV systems.

To analytically verify the latency-accuracy trade-off in FL-based IoV, we consider two core theoretical insights: (i) Impact of Computation Time on Accuracy, (ii) Impact of Communication Time on Accuracy.

As shown in the optimization convergence proofs [29], delayed updates increase gradient bias, leading to suboptimal convergence. If $T_{\text{comp},i}$ is large, AVs participate less frequently or with delayed updates, thus reducing A_g. High communication latency causes longer FL rounds and reduces the effective number of global updates within a training budget. Let R denote the total number of rounds of global aggregation. Given a convergence rate of $O\left(1/\sqrt{R}\right)$ [30], a smaller number of rounds due to higher latency results in slower accuracy improvements. This illustrates how communication inefficiency directly affects the convergence behavior and final performance of the global model in FL systems. Let ΔA denote the accuracy degradation per unit of time delay. Then, the expected accuracy loss due to latency is approximately:

$$\Delta A_g \propto \sum_{i=1}^{N} \left(\Delta A_{\text{comp},i} + \Delta A_{\text{comm},i}\right) \quad (24)$$

where $\Delta A_{\text{comp},i} = \lambda_1.T_{\text{comp},i}$ and $\Delta A_{\text{comm},i} = \lambda_2.T_{\text{comm},i}$. λ_1 and λ_2 are empirically or theoretically derived sensitivity coefficients. Thus, reducing both $T_{\text{comp},i}$ and $T_{\text{comm},i}$ directly improves the convergence and accuracy of the global model. Optimization strategies such as dynamic frequency scaling, selective AV participation, and compression of model updates are used to balance this trade-off.

IV. Experimental Scenarios and Result Analysis

This study presents a comparative analysis of FL performance under four types of model poisoning attacks, Gaussian noise, directional drift, model reversal, and random replacement, as described in Algorithm 2. In this section, we analyze the impact of varying the number of Zombie vehicles on FL performance, demonstrating how an increased proportion of

TABLE I

SIMULATION SETUP SUMMARY

Component	Details
Simulation Type	FL in IoV for Autonomous Vehicles
Vehicles	100 vehicles per scenario (with 20% to 60% designated as zombie vehicles)
FL Framework	Flower (PyTorch backend)
Defense Modules	Digital Generative Twins (DT + Gen-AI)
Attack Types	DD, GN, MR, RR
Dataset	CICIoV2024
Data Split	70% training, 10% validation, 20% testing
Evaluation Metrics	F1 Score, Accuracy, RMSE
Rounds per Scenario	5 Rounds
Noise Scale	Ranges from 0.2 to 2
Gen-AI	cGAN

compromised clients degrades model accuracy and convergence. We also highlight how the DT and Gen-AI components improve the accuracy of detection and model recovery, emphasizing their critical role in strengthening FL-based security. To ensure reproducibility and transparency, all experiments were implemented using the Flower FL framework with a PyTorch backend, and each AV, DT, and cloud aggregator was simulated on independent VMs running Ubuntu 22.04. The CICIoV2024 dataset, specifically designed for IoV environments, was used for both local and global training. Each record in the dataset includes eight features (DATA 0–DATA 7) representing the transmitted payload, an ID indicating the message priority and type, and three labeling fields, label, category and specific class, which identify whether the traffic is benign or malicious and specify its attack category and class. To simulate the heterogeneity of real-world vehicle data, the dataset for each vehicle was partitioned from CICIoV2024 [31]. Data were split into 70% training, 10% validation, and 20% testing subsets. For fair comparison, all FL experiments were carried out in 5 global rounds per scenario, with 100 total vehicles, of which 20%–60% were compromised (Zombie vehicles). The model training employed a learning rate of 0.001, a batch size of 32, epoch 3, and a binary loss of cross-entropy over five communication rounds. The detailed simulation setup is summarized in Table I. To evaluate robustness, we compared several aggregation algorithms, including FedAvg, FedAdam, Krum, and Trimmed Mean, demonstrating how robust aggregation can mitigate poisoning impacts. The F1 score was used as the main evaluation metric in all scenarios and communication rounds, as it effectively captures the trade-off between correctly detected attacks (TP), false positives (FP), and false negatives (FN), providing a reliable measure of the robustness of the model under adversarial conditions.

$$\text{F1 Score} = \frac{2 \times TP}{(2 \times TP) + FP + FN} \tag{25}$$

This study conducts eight different simulation scenarios. The following is a detailed breakdown of each scenario and the associated findings.

A. Scenario 1: 20% Zombie

Fig. 3a reveals different behavioral patterns under four attack strategies in an FL scenario with four AV and one Zombie. The greatest degradation occurs with directional drift and model reversal, where the F1 score drops sharply from 0.95 to 0.25 by Round 2 and remains suppressed, indicating that these attacks inject coherent adversarial updates that successfully mislead the global model early in training. This sudden drop suggests that the poisoned parameters were integrated into the global model after Round 1, leading to immediate performance collapse due to the absence of robust aggregation defenses. In contrast, Gaussian noise results in moderate degradation to 0.6 but stabilizes, reflecting its non-directional, stochastic nature, which can be partially averaged out by benign AV updates. Random replacement has minimal impact, with F1 scores remaining near 0.9, as the entropy introduced lacks the directional force necessary to consistently degrade model performance. In this figure, the wider bands for drift and reversal indicate higher variability and attack instability, while narrower bands indicate more consistent effects.

B. Scenario 2: 30% Zombie

In Fig. 3b, the FL scenario with 30% zombies demonstrates how structured attacks can still be effective even when benign AVs are in the majority. Directional drift reduces the F1 score to near zero in all rounds, indicating persistent and successful model corruption. The model reversal maintains high but slightly degraded F1 values, ranging from 0.8613 to 0.9352, showing partial resilience against inverse update patterns. Gaussian noise begins at 0.3098 and peaks at 0.6279 in Round 4, reflecting unstable but recoverable behavior under random perturbations. Random replacement shows significant variability, rising from 0.3838 in Round 1 to 0.8172 in Round 2, but later dropping to 0.5245 before ending at 0.6977. These results indicate that while a higher number of AVs can mitigate unstructured attacks, they are insufficient against directional drift, which still causes complete collapse.

C. Scenario 3: 40% Zombie

In Fig. 3c, the effects of adversarial strategies are analyzed in an FL scenario where 40% of the total AV are zomobies. Directional drift causes an immediate and persistent collapse in performance, with F1 scores remaining near 0.0001 throughout all rounds, indicating total model poisoning due to coherent parameter manipulation. The reverse of the model leads to

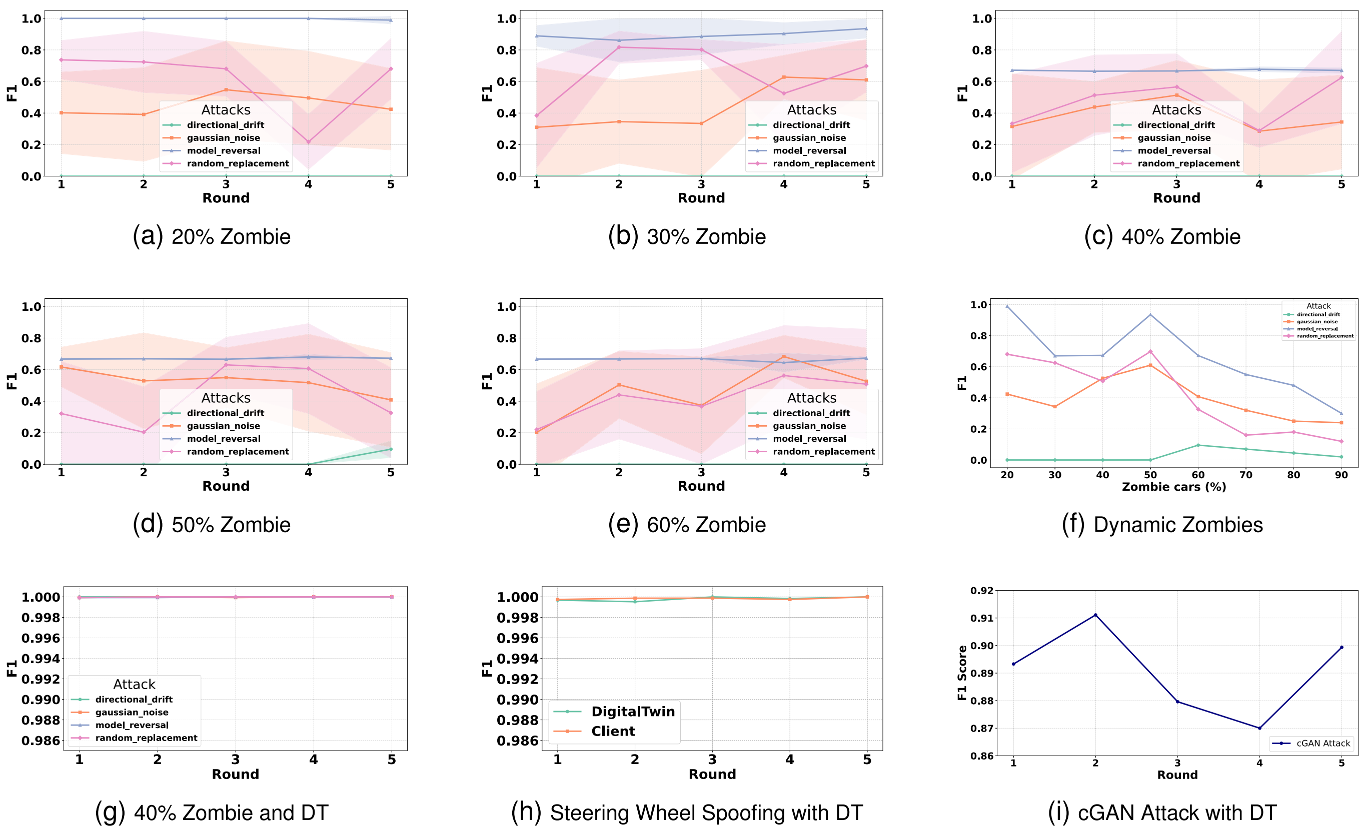

Fig. 3. Experimental results based on different scenarios including DT and GenAI.

stable but degraded performance, with F1 scores ranging from 0.6651 to 0.6775, reflecting its consistent, yet less severe impact. Gaussian noise yields highly unstable behavior, starting at 0.3148 in Round 1, peaking at 0.5134 in Round 3, then dropping sharply to 0.2853 in Round 4, highlighting the unpredictability of unstructured noise. Random replacement shows increasing volatility, improving from 0.3332 in Round 1 to 0.6251 in Round 5, but with a sharp dip to 0.2889 in Round 4, indicating inconsistent influence.

D. Scenario 4: 50% Zombie

In Fig. 3d, the FL configuration with the same number of AV and zombies exposes the vulnerability of the system to adversarial attacks. Directional drift forces the F1 score to near-zero values in the first four rounds, with only a minor increase to 0.0955 in Round 5, indicating near-total model collapse under sustained coherent manipulation. The model reversal maintains F1 scores in a tight band between 0.6658 and 0.6799, showing a consistent but moderate performance degradation. Gaussian noise starts at 0.6164, gradually declines to 0.4077, and remains unstable throughout, reflecting the unpredictable impact of non-targeted perturbations in a balanced AV composition. Random replacement exhibits high volatility, dropping from 0.3212 to 0.2031, recovering to 0.6299 in Round 3, and then again falling to 0.3257.

E. Scenario 5: 60% Zombie

In Fig. 3e, the FL setup with a number of zombies higher than the number of AVs highlights the severe impact of the increased adversarial ratio. Directional drift immediately drops the F1 score to nearly zero in all rounds, with values fluctuating only between 0.0000 and 0.0001, indicating total and sustained model collapse due to targeted manipulation. The reversal of the model produces consistently degraded results, maintaining F1 scores between 0.6444 and 0.6730, showing a stable but significant performance loss. Gaussian noise causes erratic performance, starting at 0.2022, peaking at 0.6824 in Round 4, then decreasing to 0.5259, emphasizing instability under random perturbations. Random replacement begins at 0.2197 and rises to 0.5621 before dipping to 0.5074, reflecting inconsistent disruption. The low proportion of benign AVs prevents a meaningful correction, allowing adversarial influence to dominate.

Table II presents the impact of different model poisoning strategies on FL performance in five scenarios and multiple training rounds. The results highlight how each type of attack degrades the learning process to varying degrees.

F. Impact of Adversarial Ratio on FL Performance

Fig. 3f illustrates the variation of the F1 score as the percentage of Zombie vehicles increases dynamically from 20% to 90% under four model poisoning strategies. As shown, the model reversal maintains the highest F1 score in lower zombie ratios (20–50%), close to 0.9, but experiences a gradual decline beyond 60% due to increasing adversarial influence. Random replacement achieves moderate stability up to 50% before deteriorating as the number of compromised vehicles grows. Gaussian noise demonstrates fluctuating performance

TABLE II

IMPACT OF MODEL POISONING TYPES ON FL PERFORMANCE ACROSS FIVE SCENARIOS

	Round 1				Round 2				Round 3				Round 4				Round 5			
	DD	GN	MR	RR	DD	GN	MR	RR	DD	GN	MR	RR	DD	GN	MR	RR	DD	GN	MR	RR
Scenario 1	0.001	0.40	1.0	0.73	0.001	0.391	0.99	0.72	0.001	0.54	0.99	0.68	0.001	0.49	0.98	0.21	0.001	0.42	0.98	0.68
Scenario 2	0.001	0.30	0.88	0.38	0.001	0.34	0.86	0.81	0.001	0.33	0.88	0.80	0.002	0.62	0.90	0.52	0.001	0.61	0.93	0.69
Scenario 3	0.000	0.31	0.67	0.33	0.001	0.43	0.66	0.51	0.001	0.51	0.66	0.56	0.002	0.28	0.67	0.28	0.000	0.34	0.67	0.62
Scenario 4	0.000	0.61	0.66	0.32	0.001	0.52	0.66	0.20	0.001	0.54	0.66	0.62	0.000	0.51	0.67	0.60	0.009	0.40	0.67	0.32
Scenario 5	0.000	0.20	0.66	0.21	0.001	0.50	0.66	0.43	0.001	0.37	0.66	0.36	0.000	0.68	0.64	0.56	0.000	0.52	0.67	0.50

Note: Directional Drift (DD), Gaussian Noise (GN), Model Reversal (MR), and Random Replacement (RR).

with F1 scores between 0.3 and 0.6, indicating sensitivity to random perturbations. Directional drift, on the other hand, shows near-zero detection capability across all ratios, reflecting its subtle manipulation that evades detection. Overall, the figure highlights that as the proportion of Zombie vehicles increases, the global model performance degrades across all attack types, emphasizing the importance of robust aggregation and anomaly detection mechanisms in FL for IoV security.

G. DT Integration and Effect

Integration of DT into the FL framework significantly fortifies the system's defense against AML attacks. Upon identifying potentially zombies through the proposed scoring system, the system isolates their parameter updates and historical data. These are then transmitted to the DT. The DT retrains the latest global model using the Zombie's local data and compares the resulting parameters against those originally submitted. If discrepancies indicative of manipulation are detected, the DT overrides the malicious updates by pushing the correct global model back to the Zombie, effectively mitigating the impact of the attack. To evaluate the effectiveness of the proposed solution, we tested it using scenarios IV-C. The results obtained presented in Fig. 3g present the impact of our proposed scoring system and DT.

H. Gen-AI Enhanced Verification

In this scenario, we used two different approaches to assess the resilience of the system by integrating Gen-AI with DT validation, using datasets that were completely unseen during model training. In the first approach, we introduced a new dataset that was completely isolated from AV training and testing. The DT used this dataset to train and evaluate a model. Once the model achieved high accuracy with these previously unseen data, the updated model parameters were sent to the FL server for aggregation. The server then generated a new global model, which was distributed to all AVs. To rigorously evaluate the generalizability of the system, we performed testing using a portion of the dataset that had never been seen by the AVs or the DT during any stage of training. AVs used this subset exclusively for testing, simulating real-world deployment where models must generalize to novel scenarios. The new datase included targeted attacks on steering mechanisms. Interestingly, the system maintained accurate F1 scores approaching 1 throughout all communication rounds (Fig. 3h).

In the second approach, we utilized cGAN to synthesize adversarial attacks in a more dynamic and intelligent manner (Algorithm 3). The cGAN was trained using clean model parameters and known attack deltas as input, along with condition vectors representing different types and strengths of attack. The generator in the cGAN learned to produce realistic attack patterns tailored to specific conditions, while the discriminator helped refine these outputs by distinguishing real attack deltas from generated ones. This enabled the creation of novel and targeted adversarial perturbations, which were then injected into the Digital Twin. The DT trained and validated the model with these synthetic attacks, and after achieving satisfactory performance, the parameters were forwarded to the FL server for further evaluation and testing by the AVs. The results show that the adversarial model based on cGAN maintained high F1 scores throughout the five rounds, ranging from 0.87 to 0.91. Despite slight fluctuations, the system remained robust and consistently resistant to attacks generated (Fig. 3i).

I. DL-Based IBM Analysis

Table III presents the performance comparison of the proposed DL-based IBM against the Variational Autoencoder (VAE) [32] in four attack scenarios under multiple FL aggregation strategies, including FedAvg, FedAdam, Krum and Trimmed Mean. Under the FedAvg strategy, the DL-based IBM maintains exceptionally high detection accuracy, ranging from 99.6% in Scenario 1 to 97.9% in Scenario 5, with a very low False Positive Rate (FPR) between 0.8% and 1.8%. Similarly, Krum achieves near-identical performance, maintaining accuracy greater than 98.6% in all scenarios with FPRs less than 1.3%, confirming its robustness against poisoning attacks. The FedAdam variant also delivers stable accuracy between 97.05%–98.01%, although with slightly higher FPR values (1.5%–2.3%) due to adaptive updates. In contrast, the Trimmed Mean method exhibits a noticeable drop in performance, with accuracy decreasing to 88.3%–89.5% and FPR increasing to 3.3%, suggesting that its fixed trimming threshold may not adapt well to dynamic IoV environments. The VAE baseline, which lacks federated aggregation and adversarial resilience, performs significantly worse, with accuracy dropping from 90.6% in Scenario 1 to 43.1% in Scenario 5 and FPR exceeding 4.5% under heavy attack conditions. These results collectively demonstrate that the DL-based IBM, when combined with robust FL aggregation strategies, provides superior accuracy and stability while minimizing false alarms. The low FPR and consistent accuracy across scenarios confirm the strong resilience and adaptability of the model in mitigating adversarial model poisoning attacks, ensuring trustworthy learning in large-scale IoV environments.

TABLE III
PERFORMANCE COMPARISON OF DL-BASED IBM AND VAE UNDER DIFFERENT AGGREGATION METHODS ACROSS FIVE SCENARIOS

Algorithm	Scenario 1		Scenario 2		Scenario 3		Scenario 4		Scenario 5	
	Acc.	FPR	Acc.	FPR	Acc.	FPR	Acc.	FPR	Acc.	FPR
DL-based IBM - FedAvg	99.60	0.8	99.01	1.0	99.08	1.2	98.45	1.5	97.90	1.8
DL-based IBM - FedAdam	98.01	1.5	97.15	1.8	97.20	2.0	97.06	2.1	97.05	2.3
DL-based IBM - Krum	99.50	0.7	99.00	0.9	99.01	1.0	98.95	1.1	98.60	1.3
DL-based IBM - Trimmed Mean	89.55	2.2	89.10	2.5	89.05	2.8	88.75	3.0	88.30	3.3
VAE	90.61	2.8	87.86	3.2	65.68	3.5	89.96	3.9	43.11	4.5

Note: Acc.: Accuracy (%), FPR: False Positive Rate (%).

V. CONCLUSION AND FUTURE WORK

In this study, we have investigated the impact of AML attacks, particularly model poisoning, on AVs that employ ML/DL models within their systems. To counter these threats, we have developed a framework that integrates a DL-based IBM as the local model in a FL architecture, a vehicle scoring system to identify suspicious vehicles, and a DT to validate model integrity and learn emerging attack patterns. Within this framework, the DT has been responsible for validating the models of AVs flagged by the proposed vehicle scoring system. If manipulated parameters have been detected, the DT has replaced the compromised model in the suspicious AV with the correct global model. The results have demonstrated a significant improvement in performance, with the F1 score reaching 99%. Furthermore, DT, in conjunction with Gen-AI, has been used to train the global model on novel attack patterns that AVs have not previously encountered. This approach has ensured an adaptive and resilient defense mechanism within IoV environments. As the results have shown, the F1 score for detecting new attacks by AVs has exceeded 87%, demonstrating a strong detection capability. In future work, we plan to address the scalability limitations associated with centralized components in our framework. Specifically, our goal is to reduce dependency on a single FL aggregator by exploring decentralized or hierarchical architectures, such as blockchain-based consensus mechanisms, to improve scalability, robustness, and fault tolerance. Likewise, as the current DT has operated as a centralized instance, we will extend this design toward a federated or distributed DT architecture, where multiple DT instances are deployed across edge servers (e.g., roadside units). These distributed DTs will collaboratively simulate, validate, and update models, thereby improving system resilience, reducing communication latency, and improving real-time decision-making. Furthermore, we plan to implement adaptive aggregation strategies that dynamically adjust based on network conditions and to strengthen vehicle trust assessment through fuzzy logic–based scoring, thus improving interpretability, reliability and security in large-scale IoV environments.

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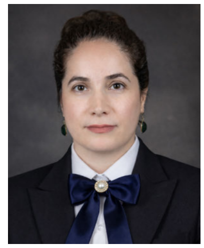

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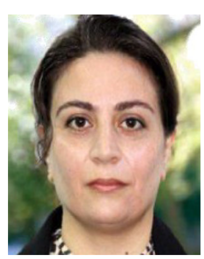

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